Exp 33:Write a program to implement the k-nearest neighbors algorithm for classification.

Code:

```
% Training data
flower(1, 1, 1, setosa).
flower(2, 2, 1, setosa).
flower(3, 1, 2, setosa).
flower(6, 5, 5, versicolor).
flower(7, 6, 5, versicolor).
flower(8, 5, 6, versicolor).

% Calculate Euclidean distance
distance(X1, Y1, X2, Y2, D) :-
   D is sqrt((X1-X2)^2 + (Y1-Y2)^2).

% Find nearest flower
nearest(X, Y, Class) :-
   findall(D-C,
      (flower(_, A, B, C),
       distance(X, Y, A, B, D)),
      List),
   keysort(List, Sorted),
   Sorted = [_-Class|_].

% KNN Classification (K = 3)
classify(X, Y, Result) :-
   findall(D-C,
      (flower(_, A, B, C),
       distance(X, Y, A, B, D)),
      List),
   keysort(List, Sorted),
   take_three(Sorted, Three),
   findall(C, member(_-C, Three), Classes),
   count_class(Classes, setosa, S),
   count_class(Classes, versicolor, V),
   (S >= V -> Result = setosa ; Result = versicolor).

take_three([A,B,C|_], [A,B,C]).
```

```
count_class([], _, 0).
count_class([X|T], X, N) :-
    count_class(T, X, M),
    N is M + 1.
count_class([X|T], Y, N) :-
    X \= Y,
    count_class(T, Y, N).
```

Output:

